ESE 3203 One-Shot Notes

Petroleum and Natural Gas Processing · KUET · past papers 2019, 2020, 2021, 2022, 2022B, 2023 · answers from the teachers' slides (Shameem sir: refining; Muhit sir: natural gas) and the textbooks they follow (Rao, Fahim et al., Sanjay Kumar, Guo & Ghalambor, Ahmed)

One-shot notes: ESE 3203 Petroleum and Natural Gas Processing

Crude oil (A1)

Fossil fuel from buried plankton/algae + heat + pressure. C 83-87, H 10-14, S 0.05-6, N 0.1-2, O 0.1-1.5 wt%; metals Ni, V; salts NaCl, $MgCl_2$, $CaCl_2$.

Families: paraffin C_nH_{2n+2} (chain), naphthene C_nH_{2n} (ring), aromatic (benzene ring); olefins only from cracking.

Classify: P/N/A/asphalt base; Mallison wax > 5% paraffinic, < 2% naphthenic; USBM key fractions 250-275 °C (atm) and 275-300 °C (40 mmHg); API > 20 conventional, < 20 heavy, < 10 bitumen; sweet < 0.5% S.

TAN = mg KOH/g (> 1 corrosive). $K_W = T_B^{1/3}/G$ (°R): 12.9 paraffinic, 10.5 naphthenic. BPSD = full-rate day; BPCD = yearly average.

Refining scheme

Separate (desalt, ADU, VDU) → convert (crack, reform, isomerize, alkylate) → treat (hydrotreat, sweeten, dewax) → blend. Complexity = conversion capacity / distillation capacity.

Energy engineer: audits, heat integration (preheat train, pinch), furnace/boiler efficiency, steam and condensate, CHP, flare gas recovery, emissions.

Fuel numbers (A2)

Octane: % iso-octane (100) in n-heptane (0) knocking the same. RON 600 rpm city, MON 900 rpm highway, sensitivity RON − MON, PON = average. Grades: regular 85-88, mid 88-90, premium > 90.

Cetane: % n-cetane (100) in α-methylnaphthalene (0) with same ignition delay; diesel 40-55. Diesel index = aniline pt (°F) × API/100.

API = 141.5/SG − 131.5: 0.802 → 44.9; 0.821 → 40.9. Flash point = lowest T vapour flashes. RVP at 37.8 °C, 4:1 air.

Octane requirement: −1 RON per 100 m altitude, per +300 rpm, per +10% humidity (0.5); +1 per +12 °C air, +6 °C coolant, 5,000-10,000 km deposits.

Pool: LSR, isomerate, reformate, FCC (38%), HC, alkylate, polymer, n-butane, oxygenates (MTBE, ETBE, TAME, ethanol: high octane, cleaner burn).

Distillation curves and desalting (A3)

TBP: 15 plates, reflux 5, atm then 40 mmHg, crude assay. ASTM D86/D1160: Engler flask, no reflux, products. EFV: flash, vapour stays with liquid, design. Sharpness TBP > ASTM > EFV.

Desalt if > 10 lb/1000 bbl: wash water 3-10% + demulsifier, heat 90-150 °C (~120), electric field 1000 V/cm (secondary) and 600 V/cm (primary), settle (Stokes). Variables: T, P, water ratio, level, injection point, demulsifier rate, water type, mixing valve Δp.

ADU and VDU (A3)

ADU: preheat 280 °C → furnace 350-400 °C → 30-50 trays, ~50 m; gas/LPG, naphtha 30-180, kerosene 180-240, diesel 240-350, residue > 350 °C; side strippers, pump-around, bottom steam.

VDU: residue would crack at > 370 °C, so 25-40 mmHg; furnace 390-450 °C; packing, wide (up to 12 m); steam ejectors (3 stages); LVGO, HVGO, lube cuts, vacuum residue. Wet/damp/dry operation.

Reforming and isomerization (A4)

Heavy naphtha RON 50-60 → reformate 90-104 + H_2. Pt 0.2-0.6% on Cl-alumina (bifunctional).

Reactions: naphthene dehydrogenation (fast, very endothermic), dehydroisomerization, paraffin dehydrocyclization (slow), isomerization (mild exo), hydrocracking (exo, bad), coke.

SR: 3-4 fixed beds + reheaters, R-1 ΔT biggest. SR 12-15 bar Pt-Re 12-24 months RON 92-100; CCR 3-10 bar Pt-Sn continuous RON 98-105, capex ×1.3.

Poisons: permanent (Pb, As, Hg, Cu, Na), coke (semi), S, N, water (temporary).

Isom.: C5-80 °C naphtha, 95-205 °C, +12-20 RON; nC5 62 → iC5 92; nC6 26 → 2,2-DMB 92-96. 2019 4c: feed 724,000 kg/h; C_3 2,520, iC_4 4,482, nC_4 12,815, C_5^+ 704,170 kg/h; isomerate RON ≈ 82.

Cracking (A5)

Thermal: heat only, free radical, olefinic.
Hydrocracking: zeolite/silica-alumina + Ni/W/Pt, high H_2 pressure, saturated S-free products, net exothermic.

Delayed coker: residue → fractionator bottom → furnace ~500 °C → coke drum (cracks there), two drums switched 16-24 h, vapours back to fractionator, coke cut by 2000-4500 psig water jets.

Visbreaking: mild, cuts viscosity/pour point. Coil 474-500 °C, quench at furnace exit; soaker 427-438 °C + soaking drum. Reactions: side chains off rings, resins → light + asphaltenes, ring cracking > 480 °C.

FCC: VGO + hot zeolite in riser 480-540 °C, 2-4 s; cyclones; spent (coked) catalyst steam-stripped → regenerator 650-815 °C burns coke → hot catalyst back (heat balance). Fixed bed = cyclic, obsolete.

Hydrotreating, H_2, lube (A6)

Feed + H_2 → heater → reactor Co-Mo/Ni-Mo on Al_2O_3, 290-400 °C, 10-100 bar → cooler → HP separator → amine scrubber, H_2 recycle → stripper.

$RSH + H_2 \rightarrow RH + H_2S$; pyridine + $5H_2 \rightarrow C_5H_{12} + NH_3$.

H_2 from reformer, steam reforming $CH_4 + H_2O \rightarrow CO + 3H_2$, shift, partial oxidation, off-gas recovery.

Lube: vacuum cuts → propane deasphalting → furfural/NMP extraction (aromatics out, VI up) → MEK dewaxing (pour pt down) → hydrofinishing (colour, stability) → additives. Lube: friction, wear, cooling, cleaning, sealing, corrosion.

Blending (A7)

Objective: meet specs and demand at least cost, max profit. Linear: SG, S, BP. Indices: $RVP^{1.25}$; flash log BI = −6.1188 + 2414/(FP − 42.6); pour $PP^{1/0.08}$.

Viscosity: BI = logv/(3 + logv), $v = 10^{3BI/(1-BI)}$.
75/100/200 cSt at 2:3:4 → BI 0.4117 → **125.8 cSt**; 72/100/201 → **124.8 cSt**.

VI = 100(L − U)/(L − H) = 100(120 − 73.3)/50 = **93.4** (table L, H give 92.4).

Natural gas and condensate (B1)

Mostly CH_4 + C_2-C_4 + N_2, CO_2, H_2S, He. Dry/wet, sweet/sour, associated/non-associated, conventional/unconventional (shale, tight, CBM, hydrates).

Origin: thermogenic (most accepted; kerogen → oil → gas with depth), biogenic (bacteria), abiogenic. Trap under cap rock.

Condensate: C_5^+ that drops out below dew point.

Inlet sep → cooling/LTS → stabilizer → splitter (LPG, naphtha, kero, gas oil).

Flare not vent: CH_4 28-34× CO_2 warming. Clean fuel: high H/C, no S, ash.

Phase envelope: bubble line, dew line, critical pt, cricondenbar (max p), cricondentherm (max T), retrograde zone. Volatile oil near T_c, GOR 2000-3300, B_o > 2.

Gas properties (B2)

pV = znRT; R = 10.73; $\gamma_g = M_a/28.97$; Kay $p_{pc} = \Sigma y p_c$. $\rho = pM/(zRT)$. $\mathbf{B_g = 0.02827zT/p}$ ft^3/scf. $c_g = 1/p - (1/z)dz/dp$.

2020: M 17.745, γ 0.6125, p_{pr} 2.96, T_{pr} 1.84, ρ 5.45, B_g 0.00858.

2021 (sour, Σy = 0.991): M 22.97, p_{pc} 680.9, T_{pc} 409.8; W-A ε 11.2 → 661.5 / 398.6; p_{pr} 2.87, T_{pr} 1.61, **z ≈ 0.83**; μ_1 0.0116 + corrections = **0.0118 cp**.

Porosity = void/bulk: 35/83 = 42.2%; 25/68 = 36.8%. Permeability = flow ability (md).

Separators (B3)

Inlet diverter (momentum), settling (gravity), mist extractor (coalescence), liquid section, LC + PC valves. LTS uses JT cooling, 0-20 °F.

Vertical: slugs, sand, small footprint; costly.
Horizontal: smaller, cheaper, high GOR, half full; poor with solids. Double barrel: liquid in lower barrel, no re-entrainment. Sphere: cheapest, little surge.
Three-phase: weir for oil, interface control for water.

De-foaming plates break foam; vortex breaker stops gas pulled into liquid outlet; centrifugal diverter; wave breakers damp sloshing.

2023 6c: ρ_g 3.845; $D^2 = 0.463/K$; V_L = 0.583 bbl.

Vertical 20 in × 6 ft; horizontal 26 in × 6.5 ft; sphere ~34 in (1/4 full; 27 in if half full).

Water and hydrates (B4)

Dehydrate: hydrates, slugging, corrosion, lower heating value; spec 6-8 lb/MMscf. Water content ↓ with p, ↑ with T, ↓ with salinity and gravity.

Campbell: W = 0.89×59 + 0.025×67 + 0.085×150 = **67 lb/MMscf**.

Hydrates: water cages + gas; need free water, low T, high p, agitation, H_2S/CO_2. Prevent: dehydrate, heat, lower p, methanol/MEG, KHI (PVP), anti-agglomerants.

Desiccant: sour → molecular sieve; cheap → alumina; 12 ppm → silica gel. Adsorption: 2 towers, down-flow drying, up-flow hot regen 230-315 °C. Glycol: TEG contactor, reboiler 175-205 °C.

Sweetening (B5)

H_2S: > 600 ppm fatal, corrosive, SO_2, catalyst poison. CO_2: no heating value, carbonic acid, freezes in LNG. Both promote hydrates.

Amine (MEA, DEA, MDEA): contactor ~38 °C high P → flash → lean/rich HX → regenerator 115-125 °C low P → acid gas to Claus.

Iron sponge: Fe_2O_3 on wood chips, < 300 ppm H_2S, pH 8-10, < 47 °C; $Fe_2O_3 + 3H_2S \rightarrow Fe_2S_3 + 3H_2O$; regen $2Fe_2S_3 + 3O_2 \rightarrow 2Fe_2O_3 + 6S$. Keeps CO_2. 3% CO_2 + 200 ppm H_2S, low budget → iron sponge.

Claus: $H_2S + 1.5O_2 \rightarrow SO_2 + H_2O$; $SO_2 + 2H_2S \rightarrow 3S + 2H_2O$.

LNG and LPG (B6)

LNG: −162 °C, 1/600 volume. Plant: inlet sep → acid gas → dehydration (< 1 ppm) → Hg removal → NGL recovery → liquefaction → N_2 rejection → storage/ship.

Cascade: propane −35, ethylene −100, methane −160 °C; each condenses the next (no single fluid spans ambient to −162).

Carrier: double hull; Moss spheres (Al, 9% Ni) or membrane (Invar); boil-off as fuel.

LPG: C_3/C_4 (BD 30/70), heavier than air. Hazards: leaks/explosion, poor storage, unattended cooking, suffocation, BLEVE. LPG Rules 2004, Explosives Act.

LPG liquid 4-15 bar; LNG −162 °C; CNG gas 200-250 bar. NGL: turbo-expander, de-ethanizer, de-propanizer, de-butanizer. Kettle: bundle under weir, vapour space above.

Reservoir flow and meters (B7)

Linear: $q = 0.001127kA\Delta p/(\mu L)$ = $0.001127\times100\times6000\times10/(2\times2000)$ = **1.69 bbl/d**; v = 0.00158 ft/d; actual v/φ = 0.0105 ft/d.

Radial: $p = 1800 + 88.28 \ln(r/0.25)$; r_e = 745 ft. Δp: 142.1, 19.7, 4.53, 0.89, 0.12 psi.

Orifice: $q = C_d A_o \sqrt{(2\Delta p/\rho(1 - \beta^4))}$, vena contracta.

Meter choice: accuracy, range, fluid, pressure loss, size, cost, output, standards.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
ADU	atmospheric distillation unit
AGA	American Gas Association
API	American Petroleum Institute
ASTM	American Society for Testing and Materials
bbl	barrel (42 US gallons, 5.615 ft^3)
BLEVE	boiling liquid expanding vapour explosion
BPCD	barrels per calendar day
BPSD	barrels per stream day
CBM	coal-bed methane
CCR	continuous catalyst regeneration
CDU	crude distillation unit
CNG	compressed natural gas
cp	centipoise
cSt	centistokes
DEA	diethanolamine
EFV	equilibrium flash vaporization
ETBE	ethyl tertiary-butyl ether
FCC	fluid catalytic cracking
FSRU	floating storage and regasification unit
GLR	gas-liquid ratio
GOR	gas-oil ratio
HDN	hydrodenitrogenation
HDS	hydrodesulfurization
HHV	higher heating value
KHI	kinetic hydrate inhibitor
LNG	liquefied natural gas
LPG	liquefied petroleum gas
LSR	light straight-run (naphtha)
LTS	low-temperature separator
md	millidarcy
MDEA	methyldiethanolamine
MEA	monoethanolamine
MEK	methyl ethyl ketone
MMscfd	million standard cubic feet per day
MON	motor octane number
MTBE	methyl tertiary-butyl ether

NGC	natural gas condensate
NGL	natural gas liquids
NMP	N-methyl pyrrolidone
PON	posted (pump) octane number
PSA	pressure swing adsorption
psia	pounds per square inch absolute
psig	pounds per square inch gauge
RON	research octane number
RVP	Reid vapour pressure
scf	standard cubic foot
SR	semi-regenerative (reformer); also straight-run
STB	stock-tank barrel
TAME	tertiary-amyl methyl ether
TAN	total acid number
TBP	true boiling point
TEG	triethylene glycol
VDU	vacuum distillation unit
VI	viscosity index